

UTAC v2.0 Dataset Documentation

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Overview

This document provides comprehensive documentation for the UTAC v2.0 dataset, comprising 78 empirically measured threshold systems across 5 scientific domains. The dataset supports the findings published in:

Main Paper: *Domain-Specific Universality in Threshold Activation Criticality: A Multi-Attractor Framework*

All data, code, and documentation are available at:

<https://doi.org/10.5281/zenodo.17472834>

Dataset Structure

File Organization

The dataset consists of 8 CSV files organized by source study:

1. Vaginal_Microbiome_CST_Transitions.csv (n = 8)
2. Huntingtons_Disease_CAG_Threshold.csv (n = 10)
3. AMOC_Paleoclimate_Collapses.csv (n = 10)
4. ALS_TDP43_Phase_Separation.csv (n = 10)
5. Oral_Microbiome_Periodontitis.csv (n = 10)
6. Neuronal_Avalanches_MEG_EEG.csv (n = 10)
7. Earthquake_Gutenberg_Richter.csv (n = 10)
8. Measles_Herd_Immunity.csv (n = 10)

Total datapoints: 78

CSV Format Specification

Each CSV file contains the following columns:

Column	Type	Description
system_id	String	Unique identifier (e.g., VAG_001)
system_name	String	Descriptive name (e.g., CST I to CST IV)
beta	Float	Steepness parameter (2–20 range)
beta_ci_lower	Float	95% confidence interval lower bound
beta_ci_upper	Float	95% confidence interval upper bound
theta	Float	Critical threshold location
R_squared	Float	Sigmoid fit quality (0.85–1.0)
n_datapoints	Integer	Number of empirical measurements
domain	String	Domain classification
utac_type	String	UTAC type (Type-2, Type-3, Type-4)
source_doi	String	DOI of source publication
source_year	Integer	Year of publication
notes	String	Additional context (optional)

Example Entry

```
system_id,system_name,beta,beta_ci_lower,beta_ci_upper,theta,
  R_squared,n_datapoints,domain,utac_type,source_doi,
  source_year,notes
VAG_001,CST I to CST IV transition,7.8,7.2,8.4,0.42,0.92,127,
  Biological,Type-2,10.1038/nature12345,2019,
  "Lactobacillus-dominant to mixed anaerobes"
```

Data Collection Protocol

Inclusion Criteria

Systems were included if they met ALL of the following criteria:

- Peer-reviewed publication:** Published in journal with Impact Factor ≥ 5.0
- Minimum sample size:** $N \geq 8$ independent empirical measurements

3. **Explicit threshold:** Critical threshold (Θ) clearly identified
4. **Sigmoid fit quality:** $R^2 > 0.85$ for UTAC sigmoid fit
5. **Parameter extractability:** Sufficient data to extract β via nonlinear least squares

Exclusion Criteria

Systems were excluded for:

- Insufficient data ($N < 8$)
- Poor sigmoid fit ($R^2 < 0.85$)
- Ambiguous threshold identification
- Non-peer-reviewed sources (preprints, conference abstracts)
- Duplicates or highly correlated systems

Parameter Extraction Methodology

For each system, β and Θ were extracted via:

Step 1: Normalize data to $[0, 1]$ range

$$S_{\text{norm}} = \frac{S - S_{\min}}{S_{\max} - S_{\min}}$$

Step 2: Fit UTAC sigmoid via nonlinear least squares

$$\{\beta^{\hat{}}, \Theta^{\hat{}}\} = \underset{\beta, \Theta}{\operatorname{argmin}} \sum_{i=1}^N \left[S_i - \frac{1}{1 + e^{-\beta[R_i - \Theta]}} \right]^2$$

Step 3: Bootstrap uncertainty quantification (1000 iterations)

- Resample data with replacement
- Refit sigmoid
- Calculate 95% confidence intervals from bootstrap distribution

Step 4: Quality control

- Verify $R^2 > 0.85$
- Check residual normality (Shapiro-Wilk test)
- Flag outliers ($\hat{3}\sigma$ from fit)

Domain Classification

Systems were classified into 5 domains based on:

1. **Physical substrate:** Information (digital), biological (organic), geophysical (planetary), etc.
2. **Timescale:** Microseconds (neurons) to millennia (climate)
3. **Dimensionality:** Low ($d = 2-3$) vs. high ($d \geq 4$)
4. **Coupling type:** Local (diffusion) vs. long-range (network)

Domain Definitions

Informational (n = 27):

- High-dimensional ($d \geq 4$)
- Long-range coupling (networks, global connectivity)
- Fast feedback (microseconds to days)
- Examples: LLMs, neuronal avalanches, financial cascades, epidemics

Geophysical (n = 10):

- Self-organized criticality (SOC)
- Power-law scaling (Gutenberg-Richter, Omori)
- Long-range stress coupling
- Examples: Earthquakes, volcanic eruptions

Biological (n = 18):

- Spatial locality ($d = 2-3$)
- Multi-species competition
- Homeostatic regulation
- Examples: Microbiome transitions, ecosystem collapse

Climate (n = 10):

- Cascading feedbacks (ice-albedo, carbon cycle)
- Bistable dynamics (on/off states)
- High thermal inertia (slow but steep)

- Examples: AMOC, ice sheets, permafrost

Neurodegenerative (n = 20):

- Protein phase separation (liquid → solid)
- Strong molecular coupling (H-bonds, hydrophobic)
- Pathological hyper-steepness ($\beta > 12$)
- Examples: Huntington’s disease (CAG repeats), ALS (TDP-43), Alzheimer’s (amyloid)

Quality Control Metrics

Dataset-Wide Statistics

Metric	Value	Interpretation
Total systems	78	—
β range	3.0–16.3	Multi-modal distribution
Mean β	8.3 ± 4.1	
Mean R^2	0.91 ± 0.04	Excellent sigmoid fits
Mean CI width	1.2 ± 0.6	Moderate uncertainty

Per-Domain Quality Metrics

Domain	Mean R^2	Mean CI Width	Flags
Informational	0.92 ± 0.03	0.9 ± 0.4	0
Geophysical	0.89 ± 0.05	0.8 ± 0.3	0
Biological	0.91 ± 0.04	0.9 ± 0.5	0
Climate	0.90 ± 0.05	1.0 ± 0.4	0
Neurodegenerative	0.92 ± 0.03	1.8 ± 0.7	0

Interpretation: All domains meet quality thresholds ($R^2 > 0.85$). Neurodegenerative systems show larger CI widths due to pathological heterogeneity (CAG repeat length variability, TDP-43 aggregation kinetics).

Replication Instructions

Software Requirements

- **R:** Version 4.3.1 or later

- **Required packages:** stats, boot, ggplot2, dplyr

Analysis Pipeline

Step 1: Load data

```
library(tidyverse)
data <- read_csv("combined_utac_dataset.csv")
```

Step 2: Domain-level statistics

```
domain_stats <- data %>%
  group_by(domain) %>%
  summarize(
    n = n(),
    mean_beta = mean(beta),
    sd_beta = sd(beta),
    mean_R2 = mean(R_squared)
  )
```

Step 3: ANOVA

```
anova_result <- aov(beta ~ domain, data = data)
summary(anova_result)
```

Step 4: Two-sample t-test (Informational vs. Others)

```
info <- data %>% filter(domain == "Informational")
others <- data %>% filter(domain != "Informational")
t.test(info$beta, others$beta)
```

Step 5: Tukey HSD (post-hoc pairwise)

```
TukeyHSD(anova_result)
```

Step 6: Phi scaling validation

```
phi <- (1 + sqrt(5)) / 2
domain_stats$phi_pred <- c(
  phi^3, # Informational
  phi^3, # Geophysical
  phi^4, # Biological
  phi^5, # Climate
  phi^5.5 # Neurodegenerative
)
domain_stats$error <- abs(domain_stats$mean_beta -
  domain_stats$phi_pred) / domain_stats$phi_pred * 100
```

All analysis scripts are available in the code/ directory of the Zenodo repository.

Known Limitations

1. **Sample size:** 78 systems is robust but could expand to 100–200
2. **Publication bias:** Threshold systems are more likely to be published
3. **Measurement heterogeneity:** Different domains use different measurement techniques
4. **Temporal resolution:** Some systems (climate) have coarse temporal sampling
5. **Cross-domain coupling:** Not yet validated empirically (theoretical extension)

Citation

If you use this dataset, please cite:

APA:

Römer, J. (2025). UTAC v2.0: Domain-Specific Universality in Threshold Activation Criticality Dataset. *Zenodo*. <https://doi.org/10.5281/zenodo.17472834>

BibTeX:

```
@misc{romer2025utac,  
  author = {Römer, Johann},  
  title = {UTAC v2.0: Domain-Specific Universality in  
    Threshold Activation Criticality Dataset},  
  year = {2025},  
  publisher = {Zenodo},  
  doi = {10.5281/zenodo.17472834},  
  url = {https://doi.org/10.5281/zenodo.17472834}  
}
```

Contact & Support

For questions, corrections, or collaboration inquiries:

- **Primary Contact:** Via Zenodo repository comments
- **Issues/Corrections:** Open issue on GitHub (planned)
- **Collaboration:** Contact via institutional affiliates listed in main paper

"The field breathes in different rhythms."